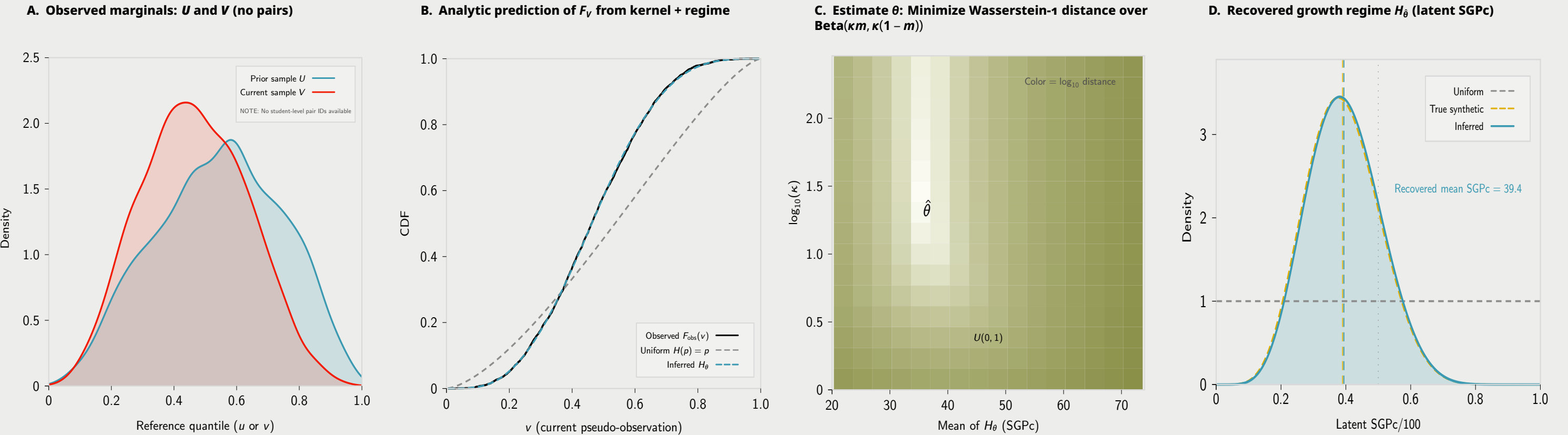


Growth as a stochastic process: From observed U and V to mean SGPC

Flow $A \rightarrow B \rightarrow C \rightarrow D \mid W_1(F_{\text{obs}}, F_\theta)$: uniform = 0.0915, inferred = 0.0019 (97.9% reduction)

Synthetic low-growth case (true mean SGPC 39), baseline t -copula kernel ($\rho = 0.72$, $\text{df} = 8$)



We observe two independent samples: prior-grade scores $\{X_i, w_i\}$ and current-grade scores $\{Y_j, w'_j\}$ for the same subgroup (e.g., a TIMSS country), but *no student-level pairs* (X_i, Y_i) are available.

We map scores to reference-percentiles using fixed reference marginals, $U = F_X^{\text{ref}}(X)$ and $V = F_Y^{\text{ref}}(Y)$, so that distributional shifts remain visible across groups and periods.

The weighted empirical distributions of U (prior; blue) and V (current; red) are therefore the only observables; all subsequent inference proceeds from these marginals plus a baseline dependence template.

Let C_0 be a baseline copula (empirical or parametric) calibrated from longitudinal data, inducing a transition kernel $F_0(v \mid u) = \Pr(V \leq v \mid U = u) = \partial C_0(u, v) / \partial u$.

We posit a *growth regime* H_θ for the latent conditional percentile $P \in (0, 1)$ (SGPC scale), with the key assumption $P \perp U$ within the subgroup.

Then the implied current-grade marginal is computed deterministically by

$$F_\theta(v) = \mathbb{E}[H_\theta(F_0(v \mid U))] \approx \frac{\sum_i w_i H_\theta(F_0(v \mid u_i))}{\sum_i w_i}.$$

Uniform growth corresponds to $H(p) = p$. Departures from uniformity tilt mass within the baseline kernel and change the implied V marginal.

We estimate θ by matching the predicted marginal F_θ to the observed F_{obs} using a distributional distance, here the Wasserstein-1 (earth mover) metric on $v \in [0, 1]$.

For a Beta regime parameterization $P \sim \text{Beta}(\kappa m, \kappa(1 - m))$ (mean m and concentration κ), we evaluate the distance on a coarse grid over (m, κ) and refine locally (e.g., L-BFGS-B).

The heatmap shows $\log_{10}$ distance: lighter regions indicate regime parameters that reproduce the observed current-grade distribution more closely. The marked point is the minimizer $\hat{\theta}$.

The fitted regime $H_{\hat{\theta}}$ (teal) is a distribution of latent conditional percentiles P (SGPC scale), i.e., how the subgroup allocates conditional ranks *within* the baseline dependence kernel $F_0(\cdot \mid u)$.

A regime centred below 50 indicates systematically lower conditional ranks relative to the reference population, while the concentration κ quantifies dispersion (a “common-cause” spread in percentile allocation).

In this synthetic validation, the recovered regime closely matches the generating regime and yields an implied V distribution that overlaps the observed current-grade marginal.